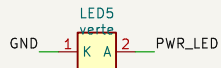
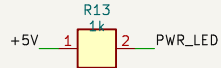
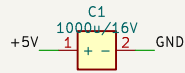
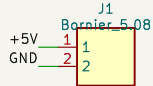
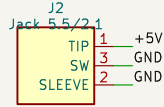
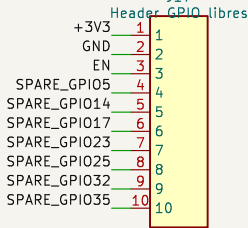


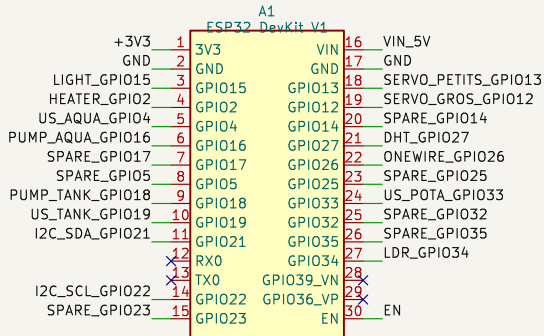
ALIMENTATION 5V (3A recommandé) – jack OU bornier.  
D5 protège le rail si l'USB du DevKit est branché en même temps.



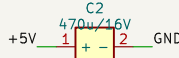
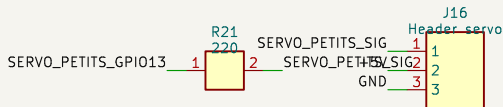
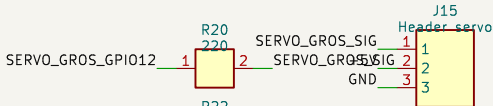
GPIO libres pour évolutions.



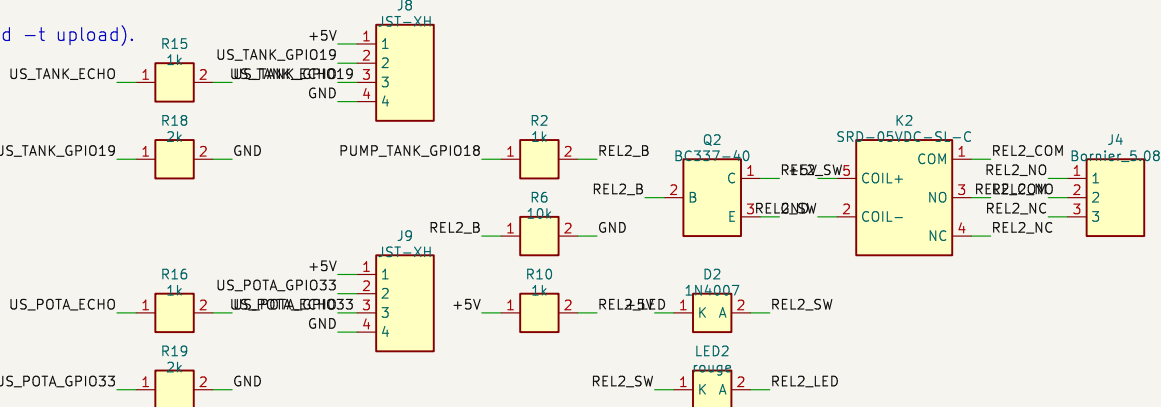
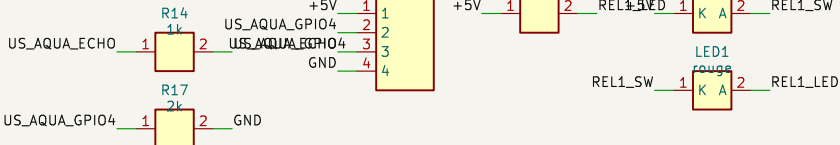
ESP32 DevKit V1 (30 broches, socketé).  
Flash/monitor via l'USB du module (pio run -e wroom-prod -t upload).



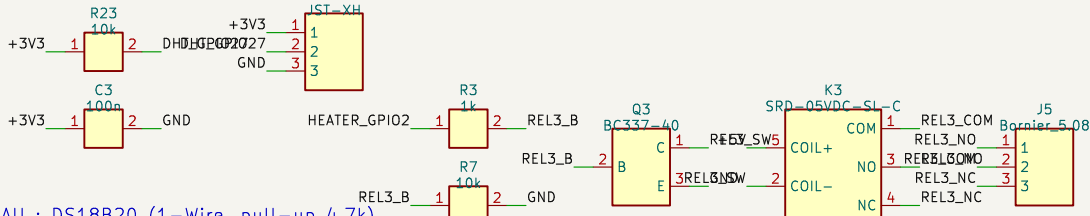
SERVOS NOURRISSEURS 5V  
GPIO12 (strapping MTDI) : pull-down R22, signal série R20.



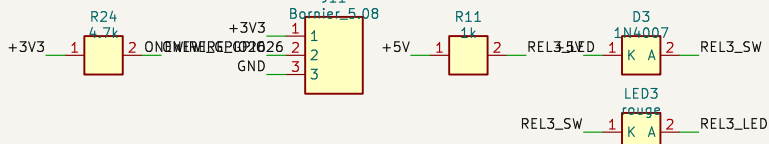
3x HC-SR04 (5V), mode mono-broche TRIG=ECHO (sensor\_ultrasonic.cpp).  
Echo 5V ramené à 3V3 par pont 1k/2k sur chaque canal.



AIR : DHT11 (option DHT22, -DUSE\_DHT22) – pull-up 10k.



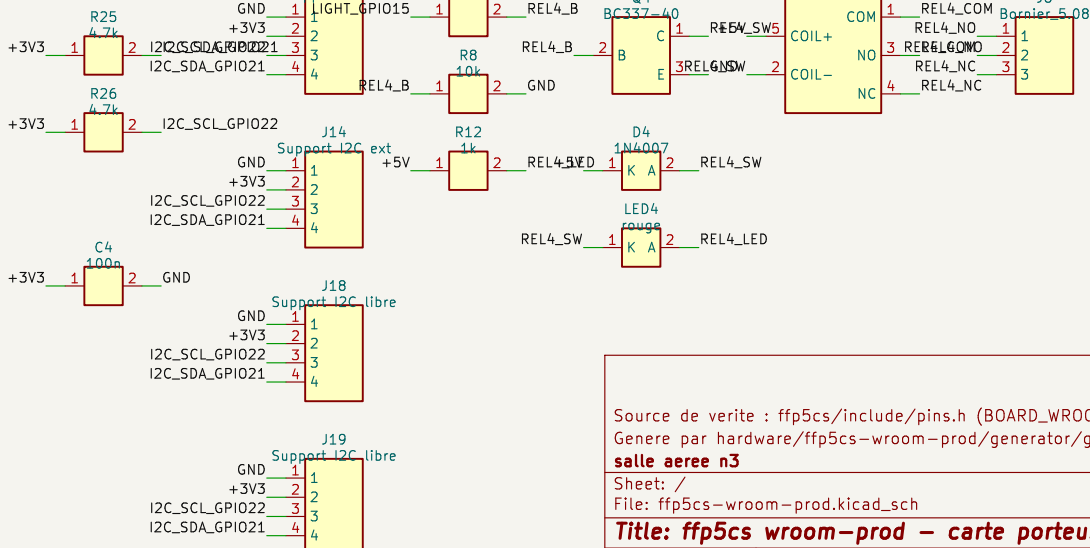
EAU : DS18B20 (1-Wire, pull-up 4.7k).



LUMIERE : LDR déportée + 10k (GPIO34 = entrée seule).



I2C : OLED SSD1306 0x3C + header extension (DS3231 si besoin).



Source de verite : ffp5cs/include/pins.h (BOARD\_WROOM) via pinmap.json  
Genere par hardware/ffp5cs-wroom-prod/generator/generate.py

salle aeree n3

Sheet: /  
File: ffp5cs-wroom-prod.kicad\_sch

Title: ffp5cs wroom-prod – carte porteuse ESP32 DevKit V1

Size: A3 Date: 2026-07-07

Rev: 0.4

KiCad E.D.A. 8.0.9

Id: 1/1